

Global Climate Change First Report: IPCC AR6 SPM WG1 Figure SPM.1 Analysis

By: Thu Phuong Duong
SID: 61444026S NTNU

March 24, 2026

This report will examine Figure SPM.1 from the IPCC Sixth Assessment Report (AR6) Working Group 1, which focuses on the history of global temperature change and the specific causes of recent warming. I choose this figure because it provides the foundational evidence for human-induced climate change by comparing long term historical reconstructions with modern atmospheric modeling.

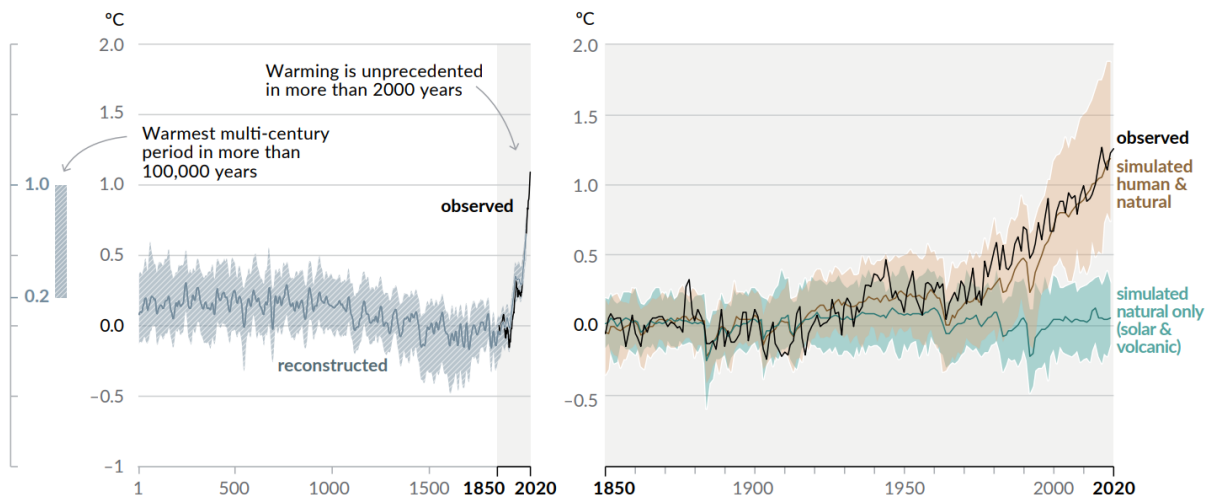


Figure 1: History of global temperature change and causes of recent warming. Panel (a) Changes in global surface temperature reconstructed from paleoclimate archives (solid grey line, years 1–2000) and from direct observations (solid black line, 1850–2020), both relative to 1850–1900 and decadal averaged. The vertical bar on the left shows the estimated temperature (very likely range) during the warmest multi-century period in at least the last 100,000 years, which occurred around 6500 years ago during the current interglacial period (Holocene). The Last Interglacial, around 125,000 years ago, is the next most recent candidate for a period of higher temperature. These past warm periods were caused by slow (multi-millennial) orbital variations. The grey shading with white diagonal lines shows the very likely ranges for the temperature reconstructions. Panel (b) Changes in global surface temperature over the past 170 years (black line) relative to 1850–1900 and annually averaged, compared to Coupled Model Intercomparison Project Phase 6 (CMIP6) climate model simulations (see Box SPM.1) of the temperature response to both human and natural drivers (brown) and to only natural drivers (solar and volcanic activity, green). Solid coloured lines show the multi-model average, and coloured shades show the very likely range of simulations.

What datasets/model/literature are used to produce these figures? What time periods/spatial areas are covered in these data? Include the links/original reference of the datasets/models at the reference list.

The figure uses a wide range of datasets. In panel (a), the authors utilize paleoclimate archives from the PAGES 2k Consortium (2019) to reconstruct the temperature for the last 2000 years. The archives consists of multi-proxy data, such as tree rings, corals, ice cores, and lake sediments. For the later era from 1850 to 2020, the authors use observational data from global datasets, such as the Berkeley Earth, HadCRUT5, NOAA-GlobalTemp, and GISTEMP. In panel (b), the authors use the results from the Coupled Model Intercomparison Project Phase 6 (CMIP6). These results consists of state-of-the-art computer simulations and are the world's most advanced climate simulations.

These data cover the entire global surface, which mean it is a spatial average of both land and ocean temperatures across the planet. In terms of time periods, the figure is divided into two distinct scales. Panel (a) covers the 2000-year period from 1 AD to 2020 AD to provide deep historical context and also provides a reference bar for the Multi-century period of the Holocene which is about 6500 years ago; while panel (b) focuses on the "Industrial Era" ranging from 1850 to 2020 to isolate the impact of human activities.

What method are used to produce these figures? (Or use plain language to explain the lines/symbols/charts in the figure).

The figure uses a method called detection and attribution. In panel (a), the authors plot the solid black line to show observed temperatures and the grey shaded area to show the margin of uncertainty. The vertical bar on the left shows that current warming is higher than anything seen in the last 100000 years. In panel (b), the authors use a comparative simulation method where on model includes both human factors (CO2 emissions, greenhouse gases, aerosols, etc.) and natural factors (sun, volcanoes, etc.), while the other one includes only the natural factors. Visually, the brown shaded region shows the combined human and natural influence, which tracks the observed black line of actual temperature rise. The green region represents climate models using only natural factors. It stays almost flat, showing that if only natural factors were at play, global temperatures would have remained nearly constant. In conclusion, the figure shows that without human activity, the Earth's temperature would have stayed relatively stable. The gap between the green and brown areas is the direct footprint of human influence.

Use your own words to explain the major findings or conclusion that can be draw from this figure. List from which section in the SMP (or Technical Summary, the full IPCC report, or any other literature) that you learn this.

The major conclusion drawn from this figure is that human influence has warmed the climate at a striking rate in the last 2000 years. The figure clearly proves that natural changes alone cannot explain the observed warming and the heat we are feeling today. Natural drivers such as solar and volcanic made a negligible contribution to total global warming since 1850. It is only when human-induced emissions are included do the models align with reality. This confirms that the warming of the atmosphere, ocean, and land is unequivocal and primarily caused by human activity.

I learned these findings in SMP, they are detailed in Section A.1 (Current State of the Climate) and are further supported by the data in Technical Summary (TS), Section 2.2, Chapter 2 of the full Physical Science Basis report.

List of Reference

- 1 **IPCC, 2021:** Summary for Policymakers. In: *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* [Masson-Delmotte, V., et al. (eds.)]. Cambridge University Press. <https://www.ipcc.ch/report/ar6/wg1/>
- 2 **PAGES 2k Consortium, 2019:** Consistent multi-decadal variability in global temperature reconstructions and simulations over the Common Era. *Nature Geoscience*, 12(8), 643-649.
- 3 **Eyring, V., et al., 2016:** Overview of the Coupled Model Intercomparison Project Phase 6 (CMIP6) experimental design and organization. *Geoscientific Model Development*, 9(5), 1937-1958.
- 4 **Kaufman et al., 2020:** A multi-method ensemble of global surface temperature reconstructions for the Holocene. *Scientific Data*, 7(1), 1-14.
- 5 **Lenssen et al., 2019:** Improvements in the GISTEMP Uncertainty Model. *Journal of Geophysical Research: Atmospheres*, 124(12), 6307-6326.